

2025 Rothschild Lecture

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1 Coordinates

Here is a QR code for a PDF of the slides:



Thank you for the invitation to speak. I am delighted to share with you all some beautiful mathematical ideas ranging over several centuries. I am most excited to also be able to talk some about how ideas in my area, the theoretical discipline of topology, are being applied to study questions related to helping people. Before we get into that, let's set the stage.

2 Bridges of Königsburg

Slide 2

In the 18th century, residents of Königsburg, Prussia played the following game: cross each of the 7 bridges of Königsburg exactly once. You could walk any way you liked through the city and around the islands, but each bridge could be crossed only once. You can imagine painting each bridge as you cross it, and then you can't cross that again.

Slide 3

If you wanted to play a more local game, you could play the “paths of Cambridge Botanical Garden”, trying to walk each of the paths exactly once. Even in framing this question, we have to make some choices: what counts as a path? How do we handle the entrances and exits? Already, we run into something I'd like us to always keep in mind: mathematics arises out of human questions and decisions, as a human endeavor.

Slide 4

Back to the 18th Century. Take a minute to try this problem on your own. I'll time you, and then I'll share Euler's answer to the problem.

Slide 5

He “Abstracted away”: instead of focusing on the particular features of the town, like the houses, the roads, the construction and size of the bridges, he winnowed down to the core features: landmasses and bridges.

He “Looked at properties of the “cartoon”: having simplified the problem by figuring out only the essential features for this, he was able to look at various features of this problem or of any like it.

He “Solved the problem there”: using the features he found, he showed no solution works in his cartoon. Since any way to cross all the bridges would solve the cartoon case, he found there is no solution.

Somehow I think that the citizens of Königsburg were less than thrilled that a mathematician had proven their attempts empty exercises!

Slide 6

Here is Euler’s approach in practice. First, divide up the land into 2 or 3 parts:

1. Starting location
2. in-transit locations
3. ending location,

and we remember that we could have ended where we started. Let’s look at the “in-transit” locations. For each of these, every time we enter, we also have to leave. This means, the bridges for this one come in pairs: in bridge and out bridge. Since we never go over the same bridge twice, we realize that each pass through uses up two bridges, and we have to have an even number of bridges connecting this land mass to the others. Now what are the bridges for each? 3, 3, 3 for the outer parts, and 5 for the inner. All odd!

Put another way, Euler turned this geometric problem into a counting problem and solved that. This is viewed by many as the start of graph theory.

Slide 7

OK. Let’s go back to the Botanical Garden, but first, a warm-up. Every Saturday, there is a 5K fun-run out in Eddington where I live. I’ve stolen the map of the path. Can we run this in such a way that we never cross the same part of the path twice? What if I start at the green flag and end at the red?

Slide 8

Let’s apply this again locally: is there a way to traverse the paths in the botanical garden once, being maximally efficient at touching grass? Take about 30 seconds to decide.

Euler looked at another classical pursuit, starting the field of topology!

3 Euler Characteristic

Slide 9

Well, as shown here, table-top gaming! If we ignore the numbers, then 5 of these are the so-called “Platonic solids”: the tetrahedron, cube, octahedron, dodecahedron, and icosahedron. These have some number of “faces” (where the numbers are written), and the number is repeated in the name in Greek, which is perhaps only helpful if you, like my sister, majored in Classics.

Two faces meet at the “edges”, and then some number of edges meet at the “vertices”.

Let’s count: Tetrahedron: 4 Faces, 6 Edges, 4 Vertices Cube: 6 Faces, 12 Edges, 8 Vertices Octahedron: 8 Faces, 12 Edges, 6 Vertices Dodecahedron: 12 Faces, 30 Edges, 20 Vertices Icosahedron: 20 Faces, 30 Edges, 12 vertices

Slide 10

What are some patterns that you notice?

1. There is some repetition: cube and octahedron swap faces and vertices, same with dodecahedron and icosahedron.
2. For all of these $F - E + V = 2$

Honestly, I never remember the number of edges in the dodecahedron or icosahedron, but I remember the duality and the numbers 12 and 20, and that gives me the rest!

Slide 11

Let’s see what happens with a non-platonic solid. The decahedron has 10 faces, 20 edges, and 12 vertices. Again, we have $F - E + V = 2$. This holds really generally.

Slide 12

This is Euler’s formula. Actually, it’s a little older than Euler: in the 16th century, a mathematician studying folding named Maurolico observed this for Platonic solids. Euler showed that in fact, it didn’t require Platonic solids. For any polyhedron, so a ball-like shape you get by gluing together polygons along their edges, you always have $F - E + V = 2$. This even holds for the “stellated” versions, like the upper left corner piece. (These are drawings by Da Vinci of various polyhedra, and the bottom left is a Keplerian model of the solar system based on the platonic solids and using the planets known at the time).

Amazingly, this also lets us answer the question: how do we know we have found all of the “regular” solids? What if we just weren’t clever enough? Euclid spent a good bit of time justifying this, but Euler’s formula can be used pretty quickly.

Slide 13

What about these? These aren't just the ball (really, the sphere) anymore. These are surfaces that have what looks like holes in them! For the outer four:

$$F - E + V = 0$$

For the inner one:

$$F - E + V = -4.$$

... I'll be honest: I didn't count these. I just used the formula:

$$F - E + V = 2 - 2 * (\# \text{ of holes}).$$

Again, this always works! We also didn't have to use straight lines and polygons, as the bottom left figure shows.

What does this buy us?

1. This formula tells us “the sphere is different than a torus, etc”. One thing we might worry about in telling surfaces apart is that, well, maybe we just aren't framing the problem in the right way. Maybe our inability to show two surfaces are the same is just a lack of imagination. Euler's formula shows that they are fundamentally different! Any way to break one up into pieces will break up the other, and, well, $2 - 2g \dots$
2. It might take us a long time to count all those faces, edges, etc, but a computer could do it almost instantly. Even better, maybe we have represented this in some non-visual way, but instead have the triangles, lines, and points and how they glue together. This is how a lot of computer graphics in old video games worked, and it is a really convenient way to represent shapes pretty easily. Even better? A computer can spit out $F - E + V$ super quickly, and this can tell us what the surface *can't* be as well as a guess about what it is.

4 Algebraic Topology

In many ways, this is the start of what we now call “algebraic topology”, the field for the semester long program “Equivariant Homotopy Theory in Context”. The basic idea is exactly what we see with Euler's formula: we attach to various spaces, shapes, etc, algebraic quantities like number of pieces, holes, etc. After Euler's early work, Poincare developed the basic tools of homotopy groups and homology, and these are now central pieces of data in modern algebra and topology. I include, because I'm giving this talk, a picture that was central in my own work with Hopkins and Ravenel solving a 50 year old problem showing that certain 254-dimensional spaces cannot exist.

If you're practically minded, you might be thinking “So what?” This sounds (and is!) very abstract, so you may be left wondering “good lord... What on Earth is this good for?” We are simple people, and 3D is good enough for us.

Maybe 4D on special occasions like birthdays or Christmas or Late May Bank Holiday. How can these high dimensional things be applied?

To data. Much of the data that we generate and analyze depends on many, many factors. Thinking of all of these potential variables means that data often naturally arises from huge spaces. And the large data sets are hard to unpack and understand. I'm going to discuss several examples of very large data sets with some really exciting payoffs in biology and then in applications to medicine and social justice.

5 TDA

Slide 14

What is this a picture of? How do you know what this is a picture of?

The image is from NASA's "Night Lights" series, and I'm going to guess that you were able to recognize it as the UK and Ireland. One of the ways we can figure this out is by squinting / varying how big our points are (or taking off glasses), you can more readily describe the underlying region. This is the big idea behind persistent homology. What are features that you can see:

1. You can roughly see the shapes of the two main land masses (there's something I won't mention in the South East, and actually some confusing stuff in the water East of Scotland)
2. You can see major population centers, along with some of the dominant roads
3. You can see the various seas and bays

What are some things we cannot see? Any parts of the UK where we don't see any sampling. Parts of the UK with a lower population density / light density are just not represented. Already, the data is showing us some social features via urban/rural.

Slide 15

Often, when we are given a data set, the data are presented in some way that falls out of the way we collected it. This might not be the most useful way to represent it! For example, if we took chairs or desks in whatever room you are sitting in, then the nature way to say where people are sitting is to talk about which chairs are filled. Put another way, the most natural way to describe locations at a performance or concert is by "row" and then "seat".

We might, however, have given the GPS coordinates of each person (as the phones in your pockets would), together with various time data to crazy precisions! Graphing all of these data points would represent all the people in the room with points in a pretty high dimensional space, and it would obscure what we actually care about: people in desks. Topological Data Analysis studies

data by trying to approximate the space it actually sits on and then figure out various features of the approximation, like the number of pieces, number of holes, Euler characteristic, etc.

How do we build these approximations? We follow the procedure that you probably used to identify the UK and the way that pointillism (like in this Seurat painting “A Sunday Afternoon on the Island of La Grande Jatte”) uses in art: lose focus / let the points grow. Mathematically, we just replace each point with a ball of some radius. We look at these all together, forgetting the individual balls anymore and just remember what region in space got filled in. (See the article [3])

Slide 16

We can vary how big we make the balls about each of our points. As these change, we get different resulting spaces. At one extreme, we see almost nothing (just the points). At the other, if the radius is too big, we also see almost nothing. In all cases, we are looking at various basic counting features:

1. How many pieces are there?
2. In each pieces, how many holes does it have?

And then the computer (or we, with paper, pencil, and a huge amount of patience) can keep going and find “higher dimensional” holes.

In these examples, at the far right, we have 1 piece and no holes. At the far left, we have 18 pieces and no holes. In the middle, we have 4 pieces, one of which has a hole. In the middle right, we have 1 piece and one hole. The middle left has 14 pieces and no holes.

Slide 17

Since those images might have been a little hard to follow, here are some directly related to the Night Light image of the UK before. I replaced (by hand, so forgive any ugliness!) each of the dots before with a very slightly bigger one. Some things stay visible: London, a fairly large midlands urban area, Cardiff Bay, and the stretch from Edinburgh to Glasgow. The Isle of Man also becomes more pronounced. Parts are still pretty sparse and others still quite disconnected

Slide 18

Here I used an even bigger radius. We have basically filled in most of Ireland, and almost all of England and Wales are now covered. If we grew much more, we would see the two landmasses merge.

What we see is that some features show up (“are born”) when we consider balls of a certain radius, and then they go away (“die”) when we reach some other critical radius. The amount of “time” (ie the stretch of radii where we see them) is measuring how long the features “persist”. This is the basis of persistent homology.

6 Explicit Applications

Butterflies

Methods of TDA have recently been applied in biology, and I'll describe some work by Andrew Blumberg and his collaborators [2]. Biologists have created trees indicating speciation, showing where various species branched off from common ancestors, traditionally using environmental and behavioral data. We can now use genetic data, pulling from large populations and seeing the spread of individual genes.

All of the butterflies here have basically the same genome, which makes it very easy to match up among species and compare. The solid lines are the tree established by lepidopterists, and the fuzzy bits are the trees given by sampling populations and analyzing their genomes. This fuzziness has been a source of confusion: was this a sampling issue or was there cross breeding early in the speciations (like with humans and neanderthals)? This horizontal gene transfer can be detected in the data: it arises as loops in the tree and these can be detected with the tools of TDA, just as we found with holes at the beginning.

Breast Cancer

One of the early successes of applying algebraic topology data to real world problems has to do with breast cancer typing. Monica Nicolau, Gunnar Carlsson, and Arnold Levine described a new subtype of breast cancer by studying the gene expressions of 295 breast cancer tumors [6]. These gave 295 data points in a 24,479-dimensional space, where each dimension measures the expression of a gene in the sample. This is huge! They found a way to reduce this down to 262-dimensions in a way that preserves the geometry, and then they applied the tools of TDA.

They found the data had a very particular shape with three clusters. These further clustered, and the researchers found a type, “c-MYB+”, which was quite clustered and for which the tumors were never metastatic and had a 100% survival rate. This type hadn't previously been identified, as it consisted of tumors that came from several distinct traditional tumor taxonomies. Imagine

Voting Enclaves

Mason Porter and collaborators have been looking at questions of how to detect pockets of different types of voters or other demographic features in an area. Bernadette Stolz, Heather Harrington, and Porter looked at Brexit voting data, building two pools of spacial data: leave and remain. These indicated distinct meaningful loops, picking out possibly expected salient political features. On the left, we see persistent loops in the “leave” votes, around London, Oxbridge, and Cardiff Bay. In remain, we see one around Birmingham.

Michelle Feng in their thesis work studied similar questions in California, and then they used a generalization of this work to study neighborhood segre-

gation as a process evolving through time, helping explain geometrically de jure segregation features as explained in books like *The Color of Law*.

COVID & Vaccination Rates

A really fantastic feature of this work is that it can be applied not only to social questions, but also to more trenchant questions like COVID-19 vaccination rates and uptakes.

Work of Abigail Hickok, Deanna Needell, and Mason Porter has looked at COVID-19 case rates over a year in neighborhoods of Los Angeles [5]. This picture combines several features into one chart:

1. The time (in days)
2. Various neighborhoods (in color)
3. the “birth” and “deaths” for various persistent features (the bottom plane)

This lets us compare the persistent features for case rates across time and neighborhood. The authors call this type of chart a “vineyard”.

Polling

Recent work at UCLA by Abigail Hickok, Benjamin Jarman, Michael Johnson, Jerry Luo, and Mason Porter uses the methods of TDA to study practical questions like access to voting sites [4]. Loosely speaking, the data points are the voting locations, and the distance used is not the “as the crow flies” distance, but rather a weighted version of how long it would take to get to the polling place by car, mass transit, or walking. This kind of chart is a “birth/death” chart, and it measures how long some particular computational feature persists. The longer it persists, the more salient it is, and in this case, that means the greater the times needed to get to some voting location. This image, taking from their upcoming paper, shows this. In Atlanta, we see that loops (and even worse, disconnected pieces) persist for longer than in Chicago. This is matching what people on-the-ground have been saying about efforts to suppress voting there by removing locations.

Geometry of Gerrymandering

The geometry of voting is deeply intertwined with the way we divide up a geographic area into distinct “districts”. In the US, the decennial census determines how many congressional seats any state gets, and then the states themselves determine how many seats their local legislatures have. Drawing the legislative maps is often (usually) the work of the state legislature, subject to complicated national and state rules. If a drawn map too heavily favors one political party, then we call it gerrymandered. Making this precise can be tricky.

The Data and Democracy Lab, organized under the direction of Moon Duchin at Cornell, looks at how we can determine if a proposed legislative map

is gerrymandered. One way to do this is to simulate valid legislative maps and compare voting outcomes. Moon Duchin, Tom Needham, and Thomas Weighill used persistent homology to compare various kinds of scale effects and to show computational features that are not just “knowing it when we see it” [1]. These kinds of questions of determining whether a legislative map is legitimate continue to influence politics in the US today.

Thank you very much for allowing me to talk about this very exciting new area, combining abstract, theoretical tools with real world problems about justice.

References

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